Note - Homework 2 - Arduino Fanning Thermometer

Design:

This is an Arduino UNO thermometer that has a 7 segment and LCD display. A servo hand fan swings according to temperature and a piezo beeps when it gets over 40 degrees. 7 segment display shows the temperature one number at a time and the LCD shows a description of the current temperature and a small animation.

Timing budget:

The system use a timer interrupt every 0.016 seconds sensor input processing, display and servo updates almost every second. It also updates a 3 state counter for different actions.

The timer ISR sets appropriate flags that are processed in the main loop.

Test results:

Everything seems to be working fine.

Known issues:

Servo can sometimes stop working and needs to be connected to 5V instead of 3.5V.

A lot of jumper wires need to be used so the design appears messy.